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KINGMODEL Dual Way Bidirectional Brushed ESC

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KINGMODEL Dual Way Bidirectional Brushed ESC Instruction Manual

Model: Dual Way Bidirectional Brushed ESC

1. Introduction

The KINGMODEL Dual Way Bidirectional Brushed Electronic Speed Controller (ESC) is designed for RC model boats and tanks utilizing 130 or 180 brushed motors. This ESC supports 2S to 3S LiPo batteries and offers both mixed and independent control modes, providing versatile control options for your remote-controlled vehicles. It features safety mechanisms such as throttle arming, over-heat protection, and throttle signal loss protection to ensure reliable operation.

2. Specifications

Feature	Specification
Output Current	Continuous 5A x 2 channels, Burst 13A (up to 10 seconds)
Input Voltage	6V-14V (2S / 3S Lithium Battery)
BEC Output	1A / 5V
Dimensions (L x W)	42mm x 26mm
Weight	17g
Recommended Motors	130, 180 Brushed Motors (Exceeding 5A per channel will damage the ESC)
Material	Composite Material
Recommended Age	14+ years

Feature	Specification
Certifications	CE, RoHS

2.1 Compatible Motor Specifications (Example: JGB37-540)

The following table provides an example of compatible motor specifications. Ensure your motor's current draw does not exceed 5A per channel for optimal performance and to prevent ESC damage.

JGB37-540

Voltage		No Load		Load Torque				Stall		Reducer		
												Weight
Workable	Rated	Speed	Current	Speed	Current	Torque	Output	Torque	Current	Ratio	Size	
Range	Volt.V	rpm	ma	rpm	A	kg. cm	W	kg. cm	A	1:00	mm	g
6-19V	12	1600	600	1120	2	1.2	25	5	5	6.25	19	260
6-19V	12	1000	600	700	2	1.8	25	7	5	10	19	260
6-19V	12	531	500	370	2	3	25	12	5	18.8	22	265
6-19V	12	333	500	266	1.8	6	25	24	5	30	22	265
6-19V	12	178	450	140	1.5	11	25	44	5	56	24	265
6-19V	12	111	400	88	1.5	18	25	72	5	90	24	265
6-19V	12	76	400	60	1.5	26	25	禁止	5	131	26.5	270
6-19V	12	59	400	47	1.5	33	25	禁止	5	168	26.5	270
6-19V	12	37	400	29	1.5	50	25	禁止	5	270	26.5	270
6-19V	12	20	400	16	1.5	60	25	禁止	5	506	29	280
6-19V	12	12	400	9	1.5	60	25	禁止	5	810	D9	280

Table: JGB37-540 Motor Specifications

3. Package Contents

- 1 x KINGMODEL Dual Way Bidirectional Brushed ESC

4. Setup and Wiring

Proper connection of the ESC to your battery, motors, and receiver is crucial for safe and correct operation. Refer to the wiring diagrams below for detailed instructions.

4.1 Connection Diagram



Figure 1: ESC Connection Overview

4.2 Simplified Wiring Diagram

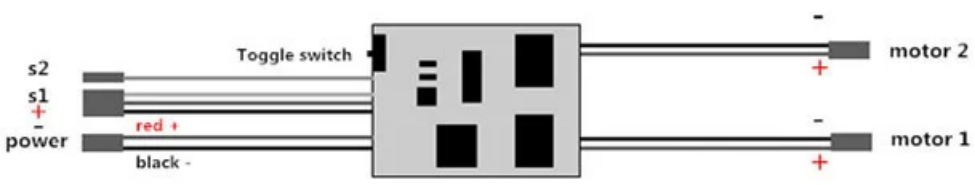


Figure 2: Simplified Wiring Diagram

4.3 Connection Steps:

1. Connect your 2S or 3S LiPo battery to the designated battery input on the ESC.
2. Connect Motor 1 to the 'Motor 1' output terminals on the ESC.
3. Connect Motor 2 to the 'Motor 2' output terminals on the ESC.
4. Connect the signal wire labeled 'Connect to 1CH/signal' to Channel 1 of your RC receiver.
5. Connect the signal wire labeled 'Connect to 2CH' to Channel 2 of your RC receiver.
6. Ensure all connections are secure and correctly polarized (red to positive, black to negative).

5. Operating Instructions

5.1 ESC Arming Procedure

To safely arm the ESC and begin operation:

1. Turn on your remote control (radio transmitter).
2. Set the S1 and S2 channels on your remote control to their median point. The ESC expects a PWM signal of approximately 1500us (with a tolerance of +/- 50us). The minimum run length is 800-1100us and the maximum is 1900-2200us, which the ESC adapts to automatically. Signals outside this range (below 800us or above 2200us) are considered illegal and will keep the ESC locked.
3. Connect the battery to the ESC. You will hear a beep from the motor, indicating power-on.
4. Once the dual channel (throttle channel) is at its midpoint (around 1500us), you will hear another "beep" from the motor, signifying that the ESC is armed and ready for operation.

5.2 Mode Switching

The ESC supports two control modes: Mixed Control and Dual Independent Control. You can switch between these modes using the physical switch on the ESC board.



Figure 3: ESC Mode Switch and LED Indicators

- **Mixed Control Mode (Default):**
  - To activate: Set the mode switch to the 'Inside' position.
  - Operation: S2 controls the throttle (100% ratio), and S1 controls the steering difference (100% ratio). Steering has priority. This mode is suitable for remote control tanks and boats.
  - LED Indicators: Green and red lights work.
- **Dual Independent Control Mode:**
  - To activate: Set the mode switch to the 'Outside' position.
  - Operation: S1 controls Motor 2, and S2 controls Motor 1. Each motor is controlled independently.
  - LED Indicators: Blue and red lights work.

6. Cautions and Safety Guidelines

- **Fail-Safe Protection:** If the signal from the remote control is lost (e.g., signal wire broken), for safety and motor protection, the throttle must be centered and unlocked again to resume operation.
- **Voltage Range:** The ESC operates within a voltage range of 6V-14V. It is compatible with 2S or 3S LiPo batteries. **Do NOT use 4S LiPo batteries or higher.**
- **Current Limit:** Each channel supports a maximum continuous current of 5A, with a burst current of 7A. Exceeding 5A per channel will likely burn out the ESC. It is recommended for 130 and 180 brushed motors.
- **Over-Heat Protection:** The ESC is equipped with over-heat protection. If the temperature of the ESC exceeds 130°C, it will gradually reduce the output power to prevent damage.
- **Throttle Signal Loss Protection:** If the throttle signal is lost for 1 second, the ESC will gradually reduce the output power. A further loss of signal for another 1 second will result in a complete cut-off of power.
- Always ensure correct polarity when connecting the battery to avoid damage to the ESC and other components.

7. Troubleshooting

Problem	Possible Cause	Solution
ESC does not arm (no motor beep)	Throttle channel not at midpoint; signal loss; incorrect wiring.	Ensure remote control throttle is centered. Check all wiring connections. Verify signal integrity.
Motors do not respond or respond erratically	Incorrect mode selected; motor current exceeds ESC limit; faulty motor/ESC.	Verify the correct control mode (mixed/independent) is selected. Check motor current draw against ESC specifications (max 5A per channel). Test with different motors if possible.
ESC gets excessively hot or burns out	Motor current draw too high; short circuit; incorrect voltage (e.g., 4S LiPo).	Ensure motor current draw is within the 5A per channel limit. Check for any short circuits in the wiring. Confirm battery voltage is 2S or 3S LiPo (6V-14V).
Loss of control during operation	Signal interference; remote control battery low; ESC signal loss protection activated.	Check remote control battery. Ensure no strong interference sources nearby. If signal is lost, re-center throttle to re-arm the ESC.

8. User Tips

- This ESC is ideal for RC model boats and tanks, providing precise speed and direction control for brushed motors.

- Always double-check your motor's current draw before connecting to prevent overloading the ESC. The 5A per channel limit is critical.
- Experiment with both mixed and independent control modes to find the best setup for your specific RC vehicle and control style.
- Ensure your remote control's channels are properly calibrated to the ESC's expected PWM signals for smooth arming and operation.

9. Warranty and Support

This product is covered by the standard manufacturer's warranty against defects in materials and workmanship. For technical support, troubleshooting assistance, or warranty claims, please contact your retailer or the manufacturer directly. Keep your proof of purchase for warranty validation.

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